Ke Tan

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About Me

Master's student in Data and Computer Science. Work on embodied AI in PyTorch, including VLA policy learning and world-modeling pipelines, and document insights in technical notes.

Education

Heidelberg University — Nov 2024 – Present
Master of Science in Data and Computer Science — *Heidelberg, Germany*
- Relevant coursework: Generative Neural Networks, Computer Vision, Reasoning with LLMs, Intelligent System

Macau University of Science and Technology — Sep 2020 – May 2024
Bachelor of Schience in Computer Science — *Macau, China*
- Ranked in the Top 5%; Best Final Year Project - Macau Peninsula Route Planning System

Projects

Reproducing RT-1-Style Vision-to-Action Policy | *PyTorch, VLA, Transformer, Robotics* — Jan 2026
- **Context**: tried to build a minimal runnable RT-1-style pipeline to connect model, training loop, and logging
- **Action**: implemented a minimal RT-1-style Transformer policy and a Behavior Cloning training loop in PyTorch to establish a runnable pipeline connecting model, training, and logging
- **Result**: documented the implementation in a public repo with detailed notes on implementation details

Reproducing Diffusion Policy for Visuomotor Control | *PyTorch, VLA, Diffusion Models, Robotics* — Jan 2026
- **Context**: tried to re-implement the training and sampling codepaths of Diffusion Policy end-to-end
- **Action**: re-implemented diffusion-based action generation in PyTorch, covering the sampling and training
- **Result**: packaged the work as a reproducible repo and summarized key technical learnings in technical notes

Edge Refinement for Light Field Disparity Estimation | *CNN, Computer Vision* — Nov 2025 – Present
- **Context**: tried adding an edge-aware refinement module to the disparity pipeline and comparing outputs to baseline
- **Action**: created an edge-aware refinement module for a disparity pipeline using CNN spatial residual feature extraction for disparity computation and residual refinement
- **Result**: achieved qualitatively sharper depth edges compared to baseline models

Work Experience

AI Application Development Intern — Mar 2024 – Aug 2024
Bihu Technology — *GuangZhou, China*
- Built an LLM-driven platform to automate teaching-material conversion via a multi-stage prompt workflow
- Designed a LangChain-based RAG learning assistant and deployed it university-wide
- **Result**: Served 5,000+ students; reviewed system architecture and authored A Survey of RAG

Data Quality Intern — Apr 2023 – Jun 2023
Yonyou Network Technology — *Macau, China*
- Validated 1M+ migrated ERP records in SQL; automated record batch generation with Python

Skills

Language: Chinese (native), English
Frameworks: PyTorch, Flask, FastAPI, SprinBoot
VLA/RL/World Models: OpenPI(Pi0/ Pi0.5), RT-1, RT-2, Diffusion Policy, OpenVLA; PPO, SAC; DINO_WM
Simulation: MuJoCo, Gazebo, ROS2, Linux-Ubuntu
Libraries: Transformers, Numpy, Pandas, SciPy, OpenCV
Programming: Python, C/C++, Docker, Git, Java, SQL